

Research Opportunities in Coastal Ecosystem Science at the University of Florida

The Cai Lab is recruiting researchers to join a new coastal Earth system science group at the University of Florida. The lab will be based at UF's Whitney Laboratory for Marine Bioscience in St. Augustine, Florida. Open positions include:

1. Postdoctoral Researcher in Numerical Modeling and Data Science

We seek a postdoctoral researcher with expertise in numerical modeling, scientific computing, and data science. Relevant experience may include hydrodynamic, reactive transport, biogeochemical, or Earth system modeling; high-performance computing; remote sensing; machine learning; and large environmental dataset analysis.

2. Postgraduate or Postdoctoral Researcher in Coastal Monitoring

We seek a researcher with experience in biogeochemical field monitoring, laboratory analysis, and environmental data analysis. Applicants with either a Ph.D. or an M.S. degree are welcome to apply. Relevant expertise may include estuarine or wetland biogeochemistry, greenhouse gas measurements, carbon and nutrient sampling, groundwater and sediment sampling, sensor deployment, and laboratory measurements.

3. Ph.D. Students in Coastal Earth System Science

Prospective Ph.D. students interested in coastal Earth system science are encouraged to apply for admission beginning in Fall 2027. Potential research topics include coastal wetlands, land-ocean interactions, carbon and nutrient cycling, hydrology, biogeochemistry, numerical modeling, and environmental data science. Students will generally be based in Gainesville, Florida, for the first one to two years, with the option to transition to the Whitney Laboratory in St. Augustine as their research develops. Depending on the project and the student's interests, the research may integrate computational approaches with field observations and measurements. Tuition and stipend support will be provided.

Interested candidates are encouraged to contact Nicole Cai at x.cai@ufl.edu with (1) CV, (2) a two-page cover letter describing research interests, relevant experience, and career goal, and (3) List of two references with contact information.

About PI:

Dr. Nicole Cai is an incoming Assistant Professor at the University of Florida, beginning in January 2027, following her NSF Postdoctoral Fellowship at Yale University. She earned her Ph.D. and master's degree from William & Mary and completed her undergraduate studies at Nanjing University. Nicole is a computational oceanographer and coastal ecosystem scientist focusing on how interacting processes across the coastal transition zone respond to and reorganize under gradual and episodic environmental stressors. She focuses on the tightly coupled physical and biogeochemical dynamics that regulate system dynamics, including inundation and flooding, saltwater intrusion, carbon cycling, tidal-wetland functions, and benthic–pelagic coupling.



More information from Nicole's personal site (<https://nicolecx122.github.io>):

About UF:

The University of Florida is a leading public research institution, offering an exceptional interdisciplinary environment and extensive research infrastructure. The Whitney Laboratory for Marine Bioscience in St. Augustine provides a distinctive coastal setting for field-based marine and estuarine research, creating opportunities to connect fundamental science with real-world environmental challenges. Both Gainesville and St. Augustine campuses offer welcoming, community-oriented environments supported by UF's campus safety, affordability, and student-support resources. Gainesville was ranked No. 5 among the best college towns in the United States for 2026, and St. Augustine was named the South's Best Small Town for 2026, offering a distinctive combination of coastal beauty, rich history, and convenient access to popular destinations such as Orlando. Researchers also benefit from HiPerGator, UF's advanced computing system, which provides extensive CPU and GPU resources, including more than 500 NVIDIA Blackwell B200 GPUs.